

UCF Physics: AST 5765/4762: (Advanced) Astronomical Data Analysis
Homework 5: Probability and Error Analysis
Due Monday, September 28, 2:30 pm

Goals: Become sufficiently familiar with probability, error analysis and fitting data to:

1. Calculate the mean, median, mode, variance, and standard deviation of a group of measurements.
2. Understand the Gaussian and Poisson distributions, their parameters, and their application in astronomical observation.
3. Remember the formulae for the basic statistics, Gaussian distribution, error propagation equation, and error arithmetic for $+$, $-$, $\times$, and $\div$. (quizzes!)
4. Fit linear models to data
5. Decide whether to believe a fit or not.

Helpful resources:

1. Chapter 6 of Bevington
2. Chapters 1, 2, 3-4.5 of Howell
3. Read the docstring of `linfit.py` (see demos)
4. AST 5765 only:
 - Read: <https://jwst-docs.stsci.edu/near-infrared-imager-and-slitless-spectrograph/> and any one (or more) of the linked pages on its four observing modes. Ambitious students may also read: <https://jwst-docs.stsci.edu/near-infrared-spectrograph>
 - Sections 14.1 – 14.3, 14.8 of Press

Remember, at all points, to explain what you do in your log and keep appropriate notes in your code! (Remember that good coding is part of your grade. Poor coding practices will lose points.)

1. **(5 points)** Use your terminal to make a new folder `hw5_<yourname>` under your `ast4762/` homework folder. For this homework, your main homework file is a Python file named `hw5_<yourname>.py` (or a `hw5_<yourname>.ipynb`, but remember to save it as a `.py` file before submission). Save it in your homework folder. Remember to commit your files and push to GitHub. Start the file with the usual header.

2. **(10 points)** Here you will continue the sigma-clipping we worked on during practicum
3. Repeat the steps from problem 2b) on the subsample you created in 2b), to make a sub-subsample.

Print the new mean, median, and standard deviation. How different are the final mean and median?

How close is the final standard deviation to that expected for the Poisson distribution?

Will this method always remove every bad pixel?

3. (10 points) Make a Python routine called `sigrej()` that carries out the “sigma rejection” process you coded during the Practicum and above (i.e., use your code from practicum problems 2b and 2c here, in a function). Note: You will use this function later in the course, so write and document it well!

The function should take as input:

- 1) an array containing your data (e.g., the array of 400 elements we made in the Practicum),
- 2) a tuple with your rejection limits (i.e., the number of standard deviations your mask will use) for each iteration (for the example we worked on before it would be (5., 5.) as in both steps we cut-off everything beyond 5σ), and
- 3) an optional Boolean mask that has the same shape as the data. The mask, if provided, indicates which data points are good (`False`=bad, `True`=good). If no mask is provided all data are initially considered good. Such a mask would be used to, e.g., label as bad the data for which the detector was faulty or the data were subsequently damaged, e.g., by a cosmic-ray hit.

The function should start by using only the data points indicated as good (`True`) in the mask.

It should then modify the mask by flagging new “bad” pixels as `False`. Note that here “bad” pixels are pixels that are further than $N\sigma$ – where N is the number provided in the tuple – away from the mean (i.e., the points that didn’t pass the rejection limit threshold).

The function should then return the modified Boolean mask.

- **Do not** use Numpy’s “masked array” feature (`numpy.ma`).

Run it on your data set from question 2 (the 400 element array). Print the mean of the “cleaned” data set and compare to your calculation from the prior step (they should be the same).

[Note: For future reference, `astropy` has a ready-made function to do sigma clipping: `astropy.sigma_clip()`, which we can use from now on.]

4. (5 points) Test that your code works on Stokes and submit your homework. If you worked in a Jupyter notebook, make sure to **save it as a .py file**. Then:

- a) Log in to Stokes. Explain how you did this in your log.
- b) Make a folder `hw5_<yourname>`. Explain how you did this in your log.
- c) Copy your `.py` file to the folder on Stokes. Explain how you did this in your log.

- d) Test it! Does it run without issues?
 - a. Yes! Great. Move on to the next steps
 - b. No. Debug!
- e) Don't forget to commit and push to GitHub. Explain what you did to do that in your log.
- f) Make a screenshot that shows your logging history, showing that you committed the file, and add it to your `hw5_<yourname>` folder. Remember to use an appropriate name for the screenshot! Don't forget to copy it to your Stokes folder, too.
- g) If needed, copy the rest of the content of your `homework/hw5_<yourname>` folder to your Stokes folder (e.g., any screenshots, etc., that might have been requested).
- h) Explain how you will do the following in your log and then copy the log into your Stokes folder:
 - a. Copy the log to the Stokes folder.
 - b. Make a tar.gz of the folder on Stokes and explain how you did this in your log.
 - c. Download the tar.gz file to your local `handin/` folder.
 - d. Submit the file to Webcourses.
- i) Copy your log into your Stokes folder and tar.gz your file.
- j) Download your tar.gz and submit your homework.